Chapter 1

Statistical Inferences for Slopes

1.1 The Regression Model and Why We Need Inference for the Slope

A sleep researcher pulls data from a sample of 8 teenagers: daily screen time, in hours, and how many hours they slept that night. She plots the points, fits a line, and finds a slope of $b_1 = -0.16$: for every extra hour of screen time, sleep drops by about ten minutes. Does this mean the effect is real across all teenagers, or could a sample this size show a slope like this by chance?

Recall from the Numerical Methods chapter that the least-squares line $\hat{y} = b_0 + b_1 x$ describes your sample: descriptive statistics, not inference.

A sample slope b_1 estimates an unknown population slope β_1, the same way $\hat{p}$ estimates p and $\bar{x}$ estimates μ. Statisticians describe the true relationship between two quantitative variables using a *population regression model*:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

β_0 is the population y-intercept, β_1 is the population slope, and ε is random error: the part of Y not explained by X. β_0 and β_1 are parameters, describing the entire population; b_0 and b_1 are statistics, calculated from your sample, used to estimate them.

The rest of this chapter reasons from b_1, the number your sample gives you, to β_1, the value you actually care about, using the same confidence interval and significance test tools you've already built for proportions and means.

Estimating the Slope and Intercept

The formulas for b_0 and b_1:

$$b_1 = r\frac{s_y}{s_x} \qquad \text{and} \qquad b_0 = \bar{y} - b_1\bar{x}$$

where r is the sample correlation coefficient and s_x, s_y are the sample standard deviations of X and Y. If you are working from raw sums instead of r, s_x, and s_y, an algebraically equivalent formula is

$$b_1 = \frac{n\sum xy - \left(\sum x\right)\left(\sum y\right)}{n\sum x^2 - \left(\sum x\right)^2}$$

Both formulas give the same b_1. You can use whichever matches the summary statistics you have.

The residual for any data point is still $e = y - \hat{y}$, where $\hat{y} = b_0 + b_1x$ is the value your line predicts. The standard deviation of all the residuals in your sample, s_e, tells you the typical size of a prediction miss. You will use s_e heavily, because it is the key in measuring how much b_1 itself might vary from sample to sample.

Following the sleep researcher's study, she samples 8 teenagers at random and records their daily screen time (X, in hours) and their sleep duration that night (Y, in hours):

Screen time (hr)	2	3	4	5	6	7	8	9
Sleep duration (hr)	7.85	8.05	7.40	7.55	6.90	7.35	6.70	6.95

For this sample, $\bar{x} = 5.5$, $\bar{y} = 7.34$, $s_x = 2.45$, $s_y = 0.47$, and $r = -0.854$. That gives

$$b_1 = (-0.854)\frac{0.47}{2.45} = -0.165 \qquad b_0 = 7.34 - (-0.165)(5.5) = 8.25$$

so $\hat{y} = 8.25 - 0.165x$. The negative slope matches what the researcher suspected: more screen time is associated with less sleep, at least in this sample. The question this chapter answers is whether that association is strong enough and consistent enough, to conclude something about β_1 for teenagers in general, or whether a slope this far from zero is a believable result of sampling variation alone.

- $Y = \beta_0 + \beta_1X + \varepsilon$ is the population regression model; β_0 and β_1 are unknown parameters, and ε is random error.
- b_0 and b_1 are sample statistics, calculated from your data, that estimate β_0 and β_1.
- $b_1 = r\frac{s_y}{s_x}$ and $b_0 = \bar{y} - b_1\bar{x}$, with an equivalent raw-sum formula available for b_1.
- The residual $e = y - \hat{y}$ and its standard deviation s_e measure how far actual data points fall from the line.

Exercise 1 Identifying the Model

A horticulturist records the amount of fertilizer applied (in grams per plant) and the resulting plant height (in centimeters) for a random sample of tomato plants.

1. Identify X and Y in this study.
2. Write the population regression model for this relationship, using correct notation.
3. In context, what would it mean if $\beta_1 = 0$?
4. Explain why β_1 is not something the horticulturist can ever know exactly, even after collecting data from every plant she owns.

Working Space

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Computing in Excel

Enter the sleep study data from earlier in this section into two columns: screen time in cells A2 through A9, and sleep duration in cells B2 through B9, with headers in row 1.

1. In an empty cell, enter `=SLOPE(B2:B9,A2:A9)`. Excel returns -0.1649, matching $b_1 = -0.165$ from the hand calculation (Excel simply keeps more decimal places).
2. In another cell, enter `=INTERCEPT(B2:B9,A2:A9)`. Excel returns 8.2506, matching $b_0 = 8.25$.
3. To check the correlation behind the calculation, enter `=CORREL(A2:A9,B2:B9)`. Excel returns -0.8536, matching $r = -0.854$.
4. To find s_x and s_y directly instead of computing them by hand, use `=STDEV.S(A2:A9)` and `=STDEV.S(B2:B9)`. These return 2.4495 and 0.4732.

Watch the argument order: for both `SLOPE` and `INTERCEPT`, the known y-values are listed first, then the known x-values. Reversing the two ranges is a common mistake, and it may hand you the wrong sign or a wrong intercept.

1.2 The Sampling Distribution of b_1 and the Confidence Interval for the Slope

Suppose the sleep researcher ran her study again with a brand new sample of 8 teenagers. A new sample of 8 teenagers would give a different fitted line and a different b_1, the same way $\hat{p}$ and $\bar{x}$ vary from sample to sample. That variation is the *sampling distribution* of b_1: the distribution of values you'd get from repeated random samples of the same size. Under the conditions for inference, this distribution is approximately normal, centered at β_1, with standard deviation σ_{b_1}.

σ_{b_1} depends on σ_ε, the population standard deviation of the errors, which is unknown, so you estimate it with s_{b_1}, the *standard error of the slope*. Substituting s_{b_1} for σ_{b_1} turns

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

from normal into a t-distribution with $df = n-2$, the same trade as with means: estimated standard error in, precision out, t-distribution in place of normal.

Unlike proportions and means, you will almost never compute s_{b_1} by hand: it depends on s_e, sample size, and the spread of your x-values, and doing that from scratch is tedious enough that no exam or workplace expects it. Instead, s_{b_1} is read directly from a calculator's regression screen or software printout, in a column usually labeled "StDev" or "SE Coef." Recognizing and interpreting that column is the actual skill.

Constructing the Confidence Interval

With s_{b_1} in hand, a confidence interval for β_1 takes the same shape as every confidence interval you have built so far: statistic plus or minus a margin of error.

$$\text{Margin of Error} = t^* s_{b_1} \qquad \text{Confidence Interval: } b_1 \pm t^* s_{b_1}$$

where t^* is the critical value from a t-distribution with $df = n-2$, chosen to match your desired confidence level.

Let's return to the sleep study. A statistical package fits the regression line to the 8 data points and reports the following output:

Predictor	Coef	StDev	T	P
Constant	8.2506	0.2448	33.70	0.000
Screen time	−0.1649	0.0411	−4.01	0.007

with $n = 8$, so $df = n - 2 = 6$. To build a 90% confidence interval for β_1, look up $t^* = t_{0.05}(6) = 1.943$. Then

$$\text{Margin of Error} = (1.943)(0.0411) = 0.0798$$

$$\text{Confidence Interval:} \ -0.1649 \pm 0.0798 = (-0.2447, -0.0850)$$

We are 90% confident that, for teenagers in general, each additional hour of daily screen time is associated with a change in average sleep duration of between -0.245 and -0.085 hours. Notice that this entire interval is negative. Since it does not contain 0, every plausible value of β_1 in this interval represents some negative relationship between screen time and sleep, which is a first hint that the effect the researcher found in her sample may reflect something real in the population, and not just sampling noise. Sections 3 and 4 make that hint precise, first by checking whether this procedure was even valid to use, and then by testing the claim directly.

- b_1 varies from sample to sample; its sampling distribution is centered at β_1 under the conditions for inference.
- The standard error s_{b_1} is read directly from calculator or computer regression output; it is rarely computed by hand.
- $t = \dfrac{b_1 - \beta_1}{s_{b_1}}$ follows a t-distribution with $df = n - 2$.
- Confidence interval: $b_1 \pm t^* s_{b_1}$, with t^* taken from the t-distribution at $df = n - 2$.
- If a confidence interval for β_1 does not contain 0, that is an early signal, not yet a conclusion, that the population slope may be nonzero.

Exercise 2 Reading Computer Output

A researcher studies the relationship between weekly study hours (X) and the improvement in exam score, in points (Y), for a random sample of $n = 12$ students. Regression output gives $b_1 = 2.35$ and $s_{b_1} = 0.68$.

1. Find the degrees of freedom for this problem.
2. Construct a 95% confidence interval for β_1. ($t^*_{0.025}(10) = 2.228$.)
3. Interpret the interval in context.

Working Space

Answer on Page ??

Computing in Excel

The regression output table above, including s_{b_1}, can be generated directly from the raw data rather than typed in by hand. Using the same screen time and sleep duration columns from Section 1 (A2:A9 and B2:B9):

1. Select a blank 2×2 range, say D2:E3. Type `=LINEST(B2:B9,A2:A9,TRUE,TRUE)` and confirm with Ctrl+Shift+Enter (in newer versions of Excel with dynamic arrays, a plain Enter is enough). This fills all four cells at once:

$b_1 = -0.1649$	$b_0 = 8.2506$
$s_{b_1} = 0.0411$	$s_{b_0} = 0.2448$

   The top row gives the coefficients, in reverse order from how you might expect (slope first, then intercept); the row directly below gives each one's standard error.

2. For the critical value, enter `=T.INV.2T(0.10,6)` in an empty cell. The first argument is $1 -$ confidence level, and the second is $df = n - 2$. Excel returns 1.9432, matching $t^*_{0.05}(6)$ from the table.

3. Margin of error: reference the two cells you already have, `=D3*[t* cell]`, giving 0.0798.

4. Confidence interval bounds: `=D2-[margin cell]` and `=D2+[margin cell]`, giving -0.2447 and -0.0850.

`LINEST` is the same idea as `SLOPE` and `INTERCEPT` from Section 1, but it returns the standard errors in the same step, so you never have to hunt for s_{b_1} separately. The argument order is the same gotcha as before: known y's first, known x's second.

1.3 Conditions for Inference on the Slope

The confidence interval you just built in Section 2 assumed that it was valid to use a t-procedure at all. Every inference method in this book comes with conditions attached, and the slope is no exception.

You have already met four of these five conditions for proportions and for means.

- **Linearity**: the true relationship between X and Y is actually linear, not curved.
- **Independence**: individual observations do not influence one another; if sampling without replacement, the population is at least 10 times the sample size.
- **Normality**: for any fixed value of X, the residuals are approximately normally distributed.
- **Equal variance**: the spread of the residuals is roughly the same across all values of X.
- **Random**: the data come from a random sample or a randomized experiment.

These are remembered by the acronym LINER. Random and Independence are conditions you check from the design of the study itself, the same way you always have. Linearity, Normality, and Equal variance are different: you check all three using the same tool, a *residual plot*, which graphs each residual against its corresponding x-value.

Checking the Conditions: The Sleep Study

Return to the sleep researcher's 8 teenagers. Before trusting the confidence interval from Section 2, check all five conditions.

Random. The researcher selected her 8 teenagers using a random sampling method, so this condition is satisfied by the design of the study.

Independence. Teenagers in general form a population far larger than 8, so sampling 8 of them without replacement does not meaningfully affect the independence of their screen time or sleep duration from one another. The 10% condition is easily satisfied.

Linearity, Normality, and Equal variance. These three come from the residuals themselves.

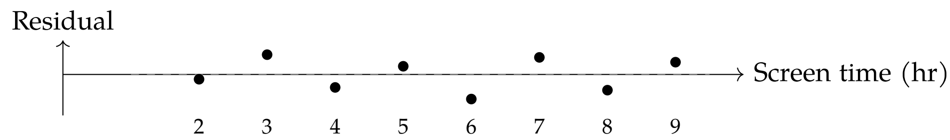

Figure 1.1: Residual plot for the sleep study regression.

The residuals in Figure 1.1 bounce above and below 0 with no visible curve, so linearity is reasonable: a straight line, rather than some curved model, is a sensible description of the relationship. The band of residuals also stays roughly the same width across the range of screen time values, from 2 hours to 9 hours, with no obvious funnel shape, so equal variance is reasonable as well.

Figure 1.2: Dotplot of the eight residuals from the sleep study regression.

The dotplot in Figure 1.2 shows no strong skew and no outliers among the eight residuals, so it is reasonable to proceed as though normality holds. With only $n = 8$ observations, though, this check is genuinely limited; a small sample simply does not give you enough information to rule out non-normality with much confidence. The honest conclusion is not "the residuals are normal," but "nothing in this small sample contradicts normality," which is a weaker, more defensible claim, and one that is worth stating explicitly on an exam or in a report.

With all five conditions reasonably satisfied, the confidence interval built in Section 2, and the hypothesis test coming in Section 4, both rest on solid ground.

- The five conditions for slope inference: Random, Linearity, Independence, Normality, Equal variance (LINER).
- Random and Independence come from the design of the study.
- Linearity, Normality, and Equal variance are checked using a residual plot (and, for normality specifically, a dotplot, boxplot, or normal probability plot of the residuals).
- A small sample size limits how confidently you can assess normality; conclusions should be stated as "nothing contradicts" rather than "confirms."

Exercise 3 Diagnosing a Residual Plot

A researcher studies the relationship between a car's weight and its fuel efficiency. The residual plot shows points that fan out widely for heavy cars but cluster tightly near 0 for light cars.

1. Which condition does this pattern call into question?
2. Explain, in your own words, why this pattern is a problem for the validity of a confidence interval or test built from this data.

Working Space

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1.4 Testing a Claim About the Slope

The confidence interval in Section 2 hinted that the true slope β_1 is not 0, since $(-0.2447, -0.0850)$ never crosses it. A hint is not a conclusion. To make the claim precise, and to attach an actual probability to the strength of the evidence, you need a significance test.

Step 1: State the Hypotheses

The sleep researcher's original question was directional: does more screen time predict *less* sleep, not just some unspecified difference. That points to a one-sided alternative.

$$H_0 : \beta_1 = 0 \qquad H_a : \beta_1 < 0$$

H_0 says screen time has no linear relationship with sleep duration in the population of teenagers; H_a says the relationship is negative. Use $\alpha = 0.05$.

Step 2: Name the Procedure and Verify Conditions

Use a one-sample t-test for the slope of a regression line, with $df = n - 2$. Section 3 already checked all five conditions, Random, Linearity, Independence, Normality, and

Equal variance, against this exact data set, and found no serious violations. Since the same data set is being reused here, that check still applies; there is no need to redo it.

Step 3: Calculate the Test Statistic and P-value

Recall the regression output from Section 2:

Predictor	Coef	StDev	T	P
Constant	8.2506	0.2448	33.70	0.000
Screen time	−0.1649	0.0411	−4.01	0.007

with $n = 8$, so $\mathrm{df} = 6$. The test statistic is

$$t = \frac{b_1 - 0}{s_{b_1}} = \frac{-0.1649}{0.0411} = -4.01$$

matching the T column exactly, since that column is always $\frac{\text{Coef}}{\text{StDev}}$.

The P column, 0.007, is a two-sided p-value: it treats $H_a : \beta_1 \neq 0$ as the alternative, whether or not that is the alternative you actually care about. Since our alternative here is one-sided, that number needs one adjustment before you can use it.

Converting a two-sided p-value to a one-sided p-value:

1. Check whether the sign of t matches the direction claimed in H_a. Positive t matches $H_a : \beta_1 > 0$; negative t matches $H_a : \beta_1 < 0$.
2. If the sign matches, divide the reported two-sided p-value by 2.
3. If the sign does not match, the data point the opposite way from your claim. The correct one-sided p-value is actually greater than 0.5, no matter how small the two-sided number looks, and there is no evidence for H_a.

Here $t = -4.01$ is negative, matching the direction of $H_a : \beta_1 < 0$, so the sign check passes. The one-sided p-value is

$$\text{p-value} = \frac{0.007}{2} = 0.0035$$

Step 4: State a Conclusion

Because $p = 0.0035 < \alpha = 0.05$, reject H_0. There is convincing statistical evidence that, in the population of teenagers this sample represents, more daily screen time is associated with a decrease in average nightly sleep duration. Because the screen time data were observed rather than randomly assigned, this conclusion is about association, not causation; some other factor could plausibly be driving both variables at once.

- Test statistic: $t = \dfrac{b_1 - 0}{s_{b_1}}$, with $df = n - 2$, using the value of β_1 specified in H_0 (usually 0).
- Standard computer output typically reports a two-sided p-value by default, regardless of which alternative you are actually testing.
- For a one-sided H_a, divide the reported two-sided p-value by 2 only if the sign of t matches the direction of H_a; otherwise the true one-sided p-value exceeds 0.5.
- Reject H_0 when $p < \alpha$; state the conclusion in context, and be careful not to claim causation from an observational study.

Computing in Excel

Continuing from the `LINEST` output in Section 2 (cell D2 for b_1, cell D3 for s_{b_1}):

1. Test statistic: in an empty cell, enter `=D2/D3`. Excel returns −4.0131, matching the T column of the regression output.
2. One-sided p-value directly: enter `=T.DIST(D2/D3,6,TRUE)`. This gives the left-tail probability $P(T < t)$ for $df = 6$, which is exactly the one-sided p-value for $H_a : \beta_1 < 0$ when t is negative. Excel returns 0.0035, with no separate halving step needed.
3. If you only have the two-sided p-value from a printed output table, use `=T.DIST.2T(ABS(D2/D3),6)/` instead, which reproduces the halving rule from Step 3 directly. Both approaches agree here: 0.0035.

`T.DIST` assumes a lower-tail test by default; for $H_a : \beta_1 > 0$, use `=1-T.DIST(t,df,TRUE)` or the equivalent `=T.DIST.RT(t,df)` instead, since you need the upper tail.

Computing in Python

The full test, from raw data to conclusion, takes only a few lines with `scipy.stats`:

```python
import numpy as np
from scipy import stats

x = np.array([2, 3, 4, 5, 6, 7, 8, 9])
y = np.array([7.85, 8.05, 7.40, 7.55, 6.90, 7.35, 6.70, 6.95])

result = stats.linregress(x, y)
df = len(x) - 2
```

```
t_stat = result.slope / result.stderr
p_one_sided = stats.t.cdf(t_stat, df)   # left tail, for Ha: beta1 < 0

print(f"b1 = {result.slope:.4f}, SE(b1) = {result.stderr:.4f}")
print(f"t = {t_stat:.4f}, df = {df}")
print(f"one-sided p-value = {p_one_sided:.4f}")
```

This prints `b1 = -0.1649, SE(b1) = 0.0411`, `t = -4.0131, df = 6`, and `one-sided p-value = 0.0035`, matching the by-hand and Excel results exactly. `stats.t.cdf(t_stat, df)` gives the lower-tail probability directly, which is the correct one-sided p-value whenever H_a is "less than and t is negative; for a "greater than" alternative, use `1 - stats.t.cdf(t_stat, df)` instead.

This is a draft chapter from the Kontinua Project. Please see our website (https://kontinua.org/) for more details.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 3)

1. X is the amount of fertilizer applied (grams per plant); Y is the resulting plant height (centimeters).
2. $Y = \beta_0 + \beta_1 X + \varepsilon$, where β_0 is the population intercept, β_1 is the population slope relating fertilizer amount to height, and ε is random error.
3. If $\beta_1 = 0$, then fertilizer amount has no linear relationship with plant height in the population; the population regression line is flat, and knowing how much fertilizer a plant received tells you nothing about how tall it will grow.
4. β_1 describes the relationship for the entire population of plants the horticulturist could ever grow, under all possible conditions, not just the ones she happened to collect data from. Even collecting data from every plant she currently owns would still leave β_1 unknown, since her plants are themselves a sample of all plants that could exist under those growing conditions. β_1 is a parameter; only a b_1 calculated from data is ever directly observed.

Answer to Exercise 3 (on page 9)

1. Equal variance. The spread of the residuals is clearly not constant across values of X; it grows as car weight increases.
2. The standard error formula used to build s_{b_1} assumes the same amount of scatter around the line at every value of X. If the scatter actually changes with X, as it does here, the standard error is not accurately estimated, so the resulting confidence interval and p-value cannot be trusted, even if every calculation is performed correctly.

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